
TASK-ALIGNED SELF-SUPERVISED LEARNING FOR MEDICAL IMAGE ANALYSIS: A SYSTEMATIC REVIEW AND PRACTICAL DESIGN GUIDELINES

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ABSTRACT

Self-supervised learning (SSL) has emerged as a promising paradigm for addressing the annotation bottleneck in medical imaging by learning representations from unlabeled data. However, the effectiveness of SSL depends heavily on the design of the pretext task and its alignment with the downstream clinical objective. In this paper, we present a systematic, task-oriented review of SSL in medical image analysis, examining how different pretext-task formulations influence performance across classification, segmentation, detection, and other clinical tasks. Following PRISMA guidelines, we analyze 75 studies published between 2017 and 2025 and organize them into four paradigms: contrastive, non-contrastive and predictive, generative and reconstruction-based, and hybrid learning. Rather than cataloguing methods by architecture, we map each paradigm to the downstream objectives it best supports. Our analysis shows that there is no universally optimal SSL strategy; instead, performance is governed by the alignment between the pretext task, the imaging modality, and the target task. Contrastive methods are effective for learning global discriminative features and align well with classification, but may overlook subtle pathological patterns. In contrast, generative and spatial prediction-based approaches better preserve local anatomical structure, making them more suitable for segmentation and other dense prediction tasks, while hybrid methods offer the most balanced performance across objectives. We further show that modality-specific design is critical and that SSL provides its greatest benefit in low-label and few-shot regimes. Finally, we distill these findings into practical design guidelines and outline key open challenges, including pathology-aware pretext task design, resource-efficient training for high-dimensional medical data, and the need for standardized evaluation protocols. This work offers practical guidance for designing more effective and clinically relevant SSL frameworks in medical imaging.

Keywords deep learning · self-supervised learning · pretext tasks · designing pretext tasks · medical image analysis

1 Introduction

The pipeline of Deep learning (DL) in medical imaging has improved and transformed diagnostic workflows, achieving expert-level performance in tasks ranging from 2D medical image analysis to volumetric medical imaging. These tasks include classification, detection, and segmentation applications, such as breast cancer classification, skin cancer detection, fracture detection, cell detection, and volumetric tumor segmentation [1, 2]. However, the performance of these tasks mainly depends on the quality of the annotations and the number of fully annotated data samples in supervised learning. Unlike the natural image domain, where annotation can be performed without expert knowledge, medical data annotation suffers from a severe “annotation bottleneck” [3]. Generating high-quality medical labels requires board-certified radiologists or pathologists, is time-consuming (ex: pixel-level segmentation of 3D Magnetic Resonance Imaging (MRI) volumes), and is subject to significant inter-observer variability [4, 5].

Furthermore, the standard paradigm of Transfer Learning pretraining models on large natural image datasets like ImageNet and fine-tuning them on medical tasks is increasingly viewed as suboptimal [6]. The domain shift between natural and medical images often results in negative transfer, where the pretrained features are irrelevant or even harmful to the target medical task [7, 8]. Therefore, the field has shifted toward Self-Supervised Learning (SSL), which extracts supervisory signals directly from the data itself, without the need for manual annotations [9]. Figure 1 illustrates the general workflow of SSL, including the pretraining stage and its adaptation to downstream tasks.

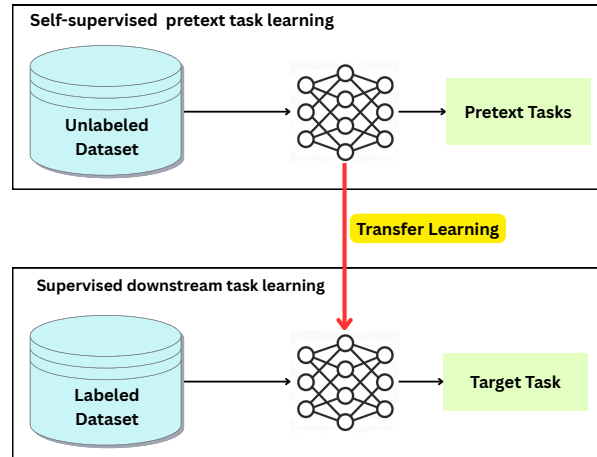


Figure 1: General workflow of SSL. The top panel illustrates the pretraining stage using unlabeled data, while the bottom panel shows adaptation to downstream tasks. Red arrows indicate the transfer learning process.

In SSL, The "pretext tasks" is used to solve side objectives like predicting missing image patches, recovering color, or contrasting augmented views. This helps to learn rich, invariant feature representations from the vast amounts of unlabeled imaging data [10]. As the field moves from Convolutional Neural Networks (CNNs) to Vision Transformers (ViTs), SSL has emerged as a central paradigm in medical Artificial Intelligence (AI). Approaches such as Masked Autoencoders (MAE) [11] and contrastive learning [12] are now setting new state-of-the-art results, particularly in low-resource settings where labeled data are limited.

Despite the rapid growth of SSL methods, there remains a gap between the design of pretext tasks and the specific requirements of downstream clinical applications. Initially, strategies of SSL in medical images often directly used general pretext tasks from computer vision, such as rotation prediction [13], jigsaw puzzle solving [14]. While these geometric transformations are effective for global classification, they may fail to capture the fine-grained, local features for dense prediction tasks [15]. For instance, enforcing invariance to rotation may improve chest X-ray classification but can be harmful for orientation-sensitive tasks in cardiac MRI. This issue, referred to as the "**alignment problem**", is that there is no "one-size-fits-all" pretext task; instead, the effectiveness of a learned representation depends on its alignment with the downstream clinical application [16].

Computer vision in the medical domain differs from general computer vision because of its inherent multimodality in data acquisition. A single patient is often scanned using multiple imaging modalities to capture complementary information. Computed Tomography (CT) reflects tissue density, MRI provides detailed soft-tissue contrast, and Positron Emission Tomography (PET) reveals metabolic activity [17]. Even within a single modality such as MRI, different sequences (T1, T2, FLAIR) offer different views of the same anatomy. This multi-modal nature creates a unique opportunity for Multimodal Self-Supervised Learning (MSSL). In natural images, multimodality typically refers to combining images with text or audio, while medical MSSL exploits the strong pixel-level correspondence across paired imaging modalities. For instance, predicting a T2-weighted MRI slice from its T1-weighted slice encourages the network to learn anatomical structures that are invariant to contrast differences [18]. Designing pretext tasks that effectively leverage this cross-modal, pixel-level relationship remains a key challenge for improving label efficiency in medical imaging [19].

This review provides a comprehensive analysis of pretext task design strictly within the domain of **medical computer vision**. We categorize SSL methods based on how their pretext tasks manipulate visual data to learn representations that are useful for downstream tasks.

The scope of this paper encompasses:

Table 1: Positioning of this review relative to representative prior surveys of SSL in medical imaging.

Survey	Year	Scope	Taxonomy by representation	Task-alignment analysis	Modality-aware discussion	Design guidelines
Shurrab & Duwairi [19]	2022	General	Partial	✗	✗	✗
Krishnan et al. [8]	2022	General	✗	✗	Partial	✗
Huang et al. [20]	2023	Classification only	Partial	Partial	Partial	✓
VanBerlo et al. [21]	2024	Radiology (X-ray/CT/MRI/US)	Partial	✗	Partial	✓
Manna et al. [22]	2024	General (descriptive)	✓	✗	Partial	✗
This review	2026	All major modalities	✓	✓	✓	✓

✓ addressed; Partial partially or implicitly addressed; ✗ not addressed.

1. Uni-modal SSL: This focuses on analyzing a single imaging type (ex: MRI or X-ray) using self-supervision to learn representations.
2. Cross-modal SSL: This focuses on analyzing from multiple imaging modalities. (ex: MRI + CT, MRI + X-rays).

We cover the main modalities such as MRI, CT, X-ray, Ultrasound, and Digital Pathology, highlighting how the characteristics of each modality guide the design of appropriate pretext tasks.

While previous surveys have successfully cataloged the architectural evolution of SSL in medical imaging, they primarily treat SSL as a generic replacement for ImageNet pre-training. However, a significant research gap remains in understanding the alignment problem: how the specific geometric and semantic invariants learned during pretext tasks, such as global semantic discrimination versus local spatial continuity, correlate with the requirements of distinct clinical objectives. Recent reviews remain narrower in scope than the present work: Huang et al. [20] provide a systematic review restricted to classification, VanBerlo et al. [21] focus on diagnostic tasks within the four radiological modalities, and Manna et al. [22] offer a comprehensive but largely descriptive account organized by methodological evolution. None of these systematically analyses the alignment between pretext task design and the requirements of distinct clinical objectives. Table 1 positions this review against representative prior surveys, highlighting that it spans all major imaging modalities and is organized around task alignment rather than architecture. Unlike existing descriptive reviews, this systematic review provides a task-centric analysis, mapping the suitability of SSL families to specific downstream tasks. Our main contributions are:

1. **A Task-Aligned Taxonomy of Pretext Tasks** - we organize existing SSL approaches into four categories, which are contrastive learning, non-contrastive and predictive learning, generative and reconstruction based learning, and hybrid learning. Unlike existing surveys, this paper is structured around the type of representations they encourage and their suitability for different downstream tasks.
2. **A Framework for Analyzing Downstream Task Alignment** - we explain a conceptual framework that examines how pretext task characteristics influence transfer performance in clinical applications such as classification, segmentation, detection, and registration. We discuss cases of both positive and negative transfer and provide evidence-based explanations.
3. **Practical Design Guidelines** - we present structured guidelines to assist in selecting and combining pretext tasks based on their target application, data regime, and modality characteristics.

This paper is organized as follows : Section 2 Background, Section 3 Systematic review methodology, Section 4 Taxonomy of pretext tasks in medical imaging, Section 5 Task alignment analysis, Section 6 Practical design guidelines for task-aligned SSL in medical imaging, Section 7 Open challenges and limitations, Section 8 Future research directions, and Section 9 Conclusion.

2 Background

2.1 Self-supervised learning paradigms

To provide a comprehensive analysis of pretext task design, we first discuss the foundational taxonomies of SSL and the specific characteristics of medical images. This background is essential for understanding the "alignment" between the pretext task objectives and the downstream clinical application.

SSL learns feature representations from unlabeled data that are useful for downstream tasks. The pretext task plays a central role in this process.

Historically, SSL methods in computer vision fall into three primary categories [19]:

1. **Predictive Learning** : These approaches apply a deterministic transformation $T(\cdot)$ to input image, such as rotation prediction, jigsaw permutation, or relative patch positioning and train the network to classify which transformation was applied [23, 24].
2. **Generative Learning** : These approaches train the network to reconstruct the original input from a corrupted or partial observation by minimizing the reconstruction loss in pixel space. Early methods used bottleneck autoencoding architectures to reconstruct inputs pixel-by-pixel [25]. More recent approaches, such as Masked Image Modeling (MIM) [11] mask a portion of the input patches and train the network to predict the masked or missing patches.
By reconstructing the masked patches, these models encourage learning global semantic context and local texture continuity without relying on specific data augmentations. In medical imaging, this method extended to predict one modality from another modality, thereby learning modality-invariant anatomical representations [18].
3. **Contrastive Learning** : These approaches operate in latent feature space (Ex: SimCLR [12], MoCo [26]). The objective is to maximize the similarity between "positive pairs" (two augmented views of the same image) while minimizing the similarity between "negative pairs" (views of different images).
The standard approach treats every image as a distinct class, learning representations that are invariant to the applied augmentations. This instance discrimination objective has been shown to produce highly transferable representations. To better support dense predication tasks such as segmentation, recent works extends to this method to pixel or patch level [27]. This method keeps the consistency between corresponding local regions across augmented views.

But, some of the new frameworks don't fit exactly into those categories because they increasingly adopt hybrid objectives that unify different SSL categories or they introduce entirely new mechanisms for representation learning [28, 29, 30]. Therefore, we divide pretext tasks into four groups: contrastive, non-contrastive, generative, and hybrid, as illustrated in Figure 2. This categorization captures recent methodological developments and provides a structured foundation for the subsequent task-alignment analysis.

2.2 Medical imaging modalities and multi-modal settings

Whereas natural images are typically 3-channel RGB projections of 3D scenes, medical images are physical measurements of biological tissues. Understanding the physics of acquisition is important for pretext task design.

- **CT** - This measures the radio-density and these are calibrated and absolute, meaning a pixel value of 0 always represents water.
- **MRI** - This measures nuclear spin relaxation times (T1, T2, PD). MRI is non-quantitative and highly susceptible to artifacts. However, MRI is multi-modal, offering naturally paired views of the same anatomy that are geometrically aligned but texturally distinct [17].
- **X-Ray and Ultrasound** - This is highly viewpoint-dependent and projective (2D). Ultrasound contains speckle noise and view-dependent artifacts.
- **Digital Pathology (WSI)** - This is gigapixel-scale images with no fixed orientation.

2.3 Downstream tasks

The use of a pretrained SSL model is evaluated on its transfer performance to downstream clinical tasks. Based on general computer vision and medical knowledge, one of the first tasks is classification, which is at the most abstract level. Dense prediction tasks, such as segmentation and detection, that preserve fine-grained texture and boundary information. Another task is registration, which involves estimating dense deformation fields and aligning scans to the same scene; similarly, reconstruction and enhancement tasks primarily focus on signal fidelity rather than semantic labels.

2.4 Alignment between pretraining and clinical tasks

The key factor in SSL for medical imaging is how well the pretext task aligns with the requirements of the downstream clinical task. Not all SSL objectives produce equally useful representations. For instance, the rotation prediction pretext task helps the model to recognize rotated images, encouraging invariance to orientation. This may be useful in pathology

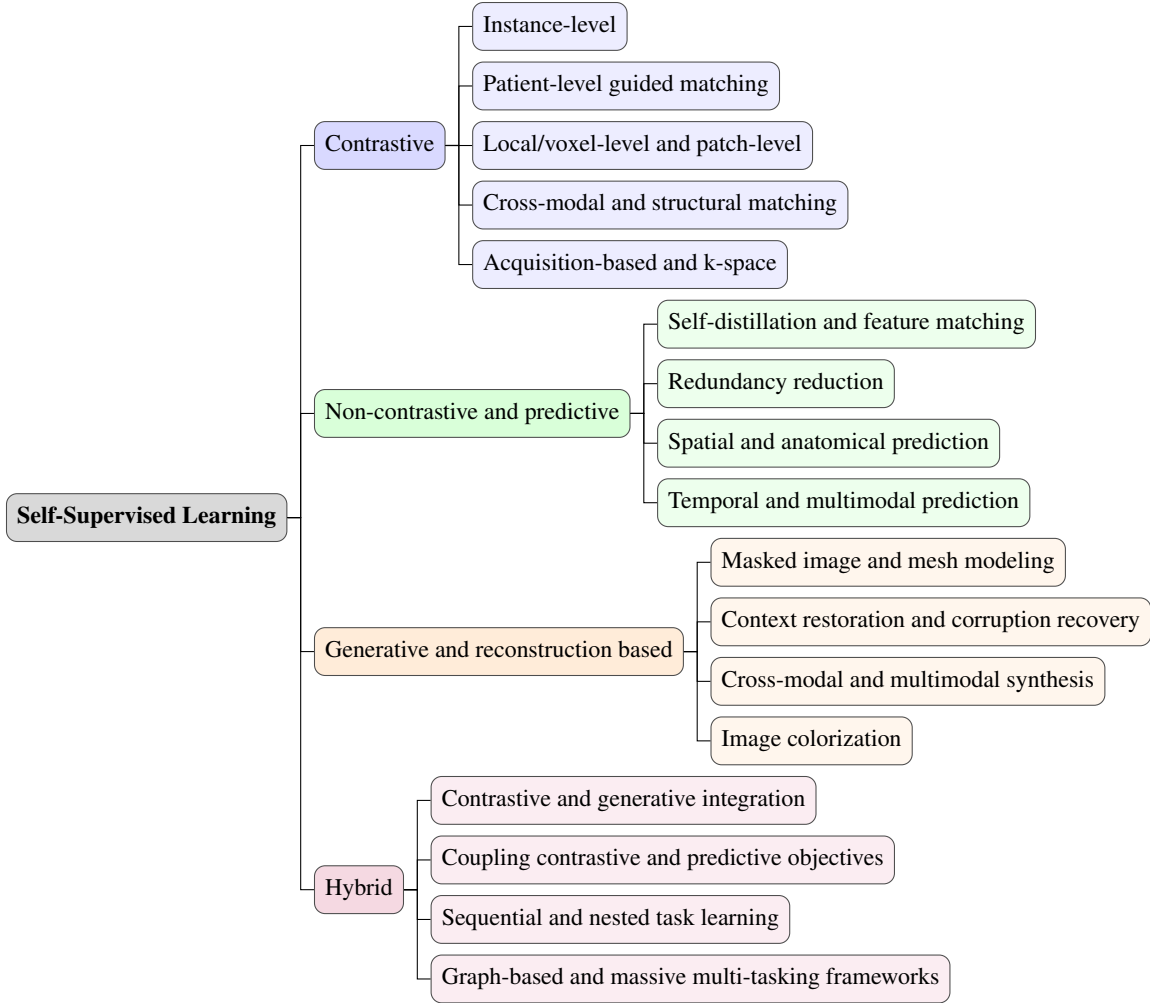


Figure 2: Literature-derived taxonomy of SSL pretext task strategies in medical image analysis.

image classification, where tissue orientation is often arbitrary. However, in cardiac MRI segmentation, anatomical orientation carries diagnostic meaning, and rotation invariance could remove clinically relevant information.

In medical imaging, the image modality also plays an important role in the choice of pretext task. For instance, colorization based pretext tasks would be situated to RGB images (Ex: WSI) but are less meaningful for grayscale modalities such as MRI. In contrast, predicting one MRI sequence from another uses known relationships between sequences and may better capture modality-specific anatomical structure.

3 Systematic review methodology

This systematic review was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guidelines [31]. The objective of the review was to identify and synthesize peer-reviewed studies that apply SSL techniques to medical image analysis tasks and to analyze how different pretext task designs align with downstream clinical objectives.

A structured literature search was performed to identify relevant studies published between **January 2017 and December 2025**. The review focused on research that proposed or evaluated SSL approaches for medical imaging applications including classification, segmentation, detection, reconstruction, and related clinical tasks. The review protocol defined the search strategy, inclusion and exclusion criteria, screening procedure, and data extraction framework prior to conducting the study selection process.

3.1 Search strategy and databases

A systematic literature search was conducted across several major academic databases to ensure comprehensive coverage of research in both medical imaging and machine learning. The primary databases included **PubMed, IEEE Xplore, Scopus, and Web of Science**, which index the majority of peer-reviewed publications in biomedical engineering, computer vision, and AI.

In addition to these databases, **Google Scholar** was used as a supplementary search engine to identify additional relevant studies, including conference papers, preprints, and interdisciplinary publications that may not yet be indexed in traditional databases.

The search covered publications **from 2017 to December 2025**, reflecting the rapid development of self-supervised learning methods in medical image analysis.

The search strategy used combinations of keywords related to self-supervised learning and medical imaging. Core methodological terms included “self-supervised learning”, “contrastive learning”, “non-contrastive learning”, “predictive learning”, and “generative learning”. These were combined with domain-specific keywords such as “medical imaging” and “medical image analysis”, as well as modality-specific terms including “CT”, “computed tomography”, “MRI”, “magnetic resonance imaging”, “X-ray”, “radiograph”, “pathology”, “histopathology”, and “whole slide image (WSI)”. Boolean operators (AND/OR) were used to construct the search queries and refine the results.

The results of the literature search and the study selection process are summarized in the PRISMA flow diagram shown in Figure 3.

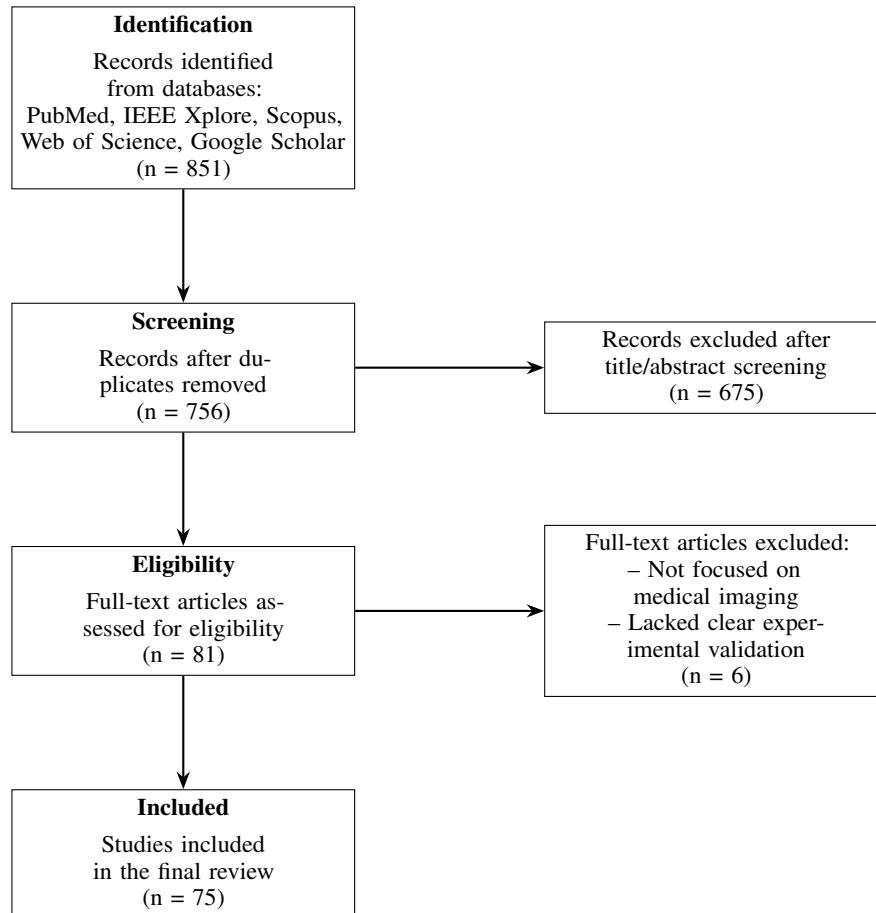


Figure 3: PRISMA flow diagram illustrating the study selection process for the systematic review.

3.2 Inclusion and exclusion criteria

We included studies that proposing a novel SSL pretext task or a novel application/adaptation of an existing task to medical data, proposed or evaluated a SSL pretraining strategy, applied the method to medical imaging data, explicitly evaluate the learned representations on at least one downstream clinically related task with quantitative metrics, and provided a clear comparison with a supervised baseline or other SSL methods. To ensure technical and clinical rigor, the selection was focused on peer-reviewed studies from reputable publishers, including **IEEE, Elsevier, Springer, Nature, Frontiers, and MDPI, as well as top-tier AI conference proceedings**. High-quality pre-prints from **arXiv** were also included if they proposed significant architectural innovations or addressed critical research gaps not yet covered in journal literature.

We excluded studies that focused only on supervised or semi-supervised learning without an SSL pretraining stage, did not use medical imaging data, lacked clear experimental validation, or were not written in English.

3.3 Study selection process

After removing duplicates, the remaining records were screened based on the relevance of their titles and abstracts to SSL in medical image analysis. 81 articles were retained for full-text assessment. Following the predefined inclusion and exclusion criteria, a total of 75 studies were included in the final analysis.

3.4 Data extraction

For all included studies, data were extracted into a structured schema to facilitate the "Task Alignment" analysis. The extraction fields included:

1. The imaging modality used and whether the setting was Uni-modal or Cross-modal.
2. We categorized pretext tasks into Contrastive, Non-contrastive learning and predictive learning, Generative and reconstruction based learning, or Hybrid Learning.
3. Specific clinical applications (ex: tumor segmentation, disease classification) were identified.
4. The evaluation metrics used to assess performance.

3.5 Review protocol and quality considerations

The review followed a protocol defined prior to study selection, specifying the search strategy, eligibility criteria, and data-extraction schema described above. The protocol was not registered in a public registry such as PROSPERO, which we note as a limitation. Title/abstract screening and full-text assessment were conducted against the predefined inclusion and exclusion criteria, and uncertain cases were resolved by re-examination of the full text.

As this review synthesises methodological trends rather than pooling clinical outcomes, a formal meta-analysis and quantitative risk-of-bias scoring were not performed. Instead, eligibility was restricted to peer-reviewed studies that reported quantitative downstream metrics and an explicit comparison against a supervised or alternative SSL baseline, which provides a baseline level of methodological rigour across the included studies. We acknowledge that the absence of a standardised quality-appraisal instrument is a limitation, and that heterogeneity in datasets, backbones, and evaluation protocols across studies precludes direct numerical comparison of reported performance figures.

4 Taxonomy of pretext tasks in medical imaging

Building on the four-family scheme motivated in Section 2, we organize the reviewed pretext tasks into Contrastive; Non-contrastive and predictive; Generative and reconstruction-based; and Hybrid learning. The first three families are defined by their dominant pretext objective, while Hybrid methods are those that explicitly optimise objectives spanning more than one family. This taxonomy is structured by the dominant learning objective and by how supervision is derived from unlabelled data, providing the foundation for the downstream task-alignment analysis in Section 5.

4.1 Contrastive learning

4.1.1 Instance-level contrastive learning

The standard contrastive formulation treats two augmented views of the same image as a positive pair, while views from different images serve as negatives. Several foundational contrastive frameworks have been directly adapted

to medical imaging. The SimCLR framework [12] creates two stochastic augmentations of an image and optimizes their representation similarity. This formulation has been widely applied in digital histopathology representation learning [32], liver view classification in ultrasound [33], and biparametric prostate MRI classification, where random patches from various sequences (T2W, DWI, ADC) were augmented with Rician noise and intensity shifts to learn robust slice-level features [34]. Similarly, discriminative contrastive learning has been used to pretrain cross-scale U-Net encoders for brain tumor segmentation in schizophrenia, aligning features of the same MRI scan against others in the latent space [35].

MoCo and its versions [26] rely on a memory bank of negative samples. These methods have been effectively used for COVID-19 prognosis from radiographs [36], few-shot COVID-19 diagnosis from CT slices [37], and chest X-ray pretraining in CheSS [38]. Expanding on this, the Nearest-Neighbor Contrastive Learning (NNCLR) framework was applied to coronal T1 MRI slices for neuro-degenerative disorder classification, using a large queue of negative samples to learn structural brain representations [39].

PaRCL [40] used the MoCo-v2 framework for glaucoma classification on fundus images, but constructed positive pairs without strong data augmentation. Instead, multi-layer feature representations from the same image were aggregated using a subnetwork ensemble strategy within query and momentum encoders.

4.1.2 Patient-level guided matching

In medical imaging, multiple images of the same patient often share pathology-specific information. Several works therefore redefine positive pairs using patient identity or metadata. The MICLe [7] created patient-level “bags” of images sharing the same pathology but collected under different views or lighting conditions. Two different images from the same bag are randomly sampled to form a positive pair.

In ophthalmology, Self-FI [41] predicted whether two augmented views originate from the same fundus image. In addition, they added bilateral matching between the left and right eyes of the same patient. The framework [42], which used longitudinal spinal MRI scans to construct positive pairs from vertebral bodies of the same patient and negative pairs from different individuals. A Siamese network trained with a contrastive loss learned identity-consistent representations by pulling same-patient embeddings together while pushing different-patient embeddings apart. Another retinal framework has done patient-level instance discrimination by matching a patient’s fundus image, an augmented fundus image, and synthesized Fundus Fluorescein Angiography, while separating representations across different patients [43].

4.1.3 Local, voxel-level, and patch-level matching

Dense prediction tasks require preservation of spatial structure, leading to localized contrastive objectives. The vox2vox [44] predicted whether two voxels from overlapping augmented 3D patches correspond to the exact same spatial location in the original volume. This voxel matching enforces anatomical consistency in volumetric data.

For 3D dental segmentation [45], dental meshes were augmented and used K-Nearest Neighbors to sample corresponding overlapping jaw regions. This study matched features across these localized regions under a contrastive objective.

Focusing on localized pathology, a framework for 3D necrotic lung lesion segmentation [46] used dependency augmentation to align representations across inter-slice and longitudinal CT slices within RECIST-cropped regions. To specifically capture the internal structure of the pathology, they used a distance-transform-based mask-out augmentation to simulate necrotic cores, enabling the model to learn slice-invariant and necrosis-robust features at a granular, patch-based level.

The study [47] combines instance-level contrastive loss on WSI patches with a clustering-based objective. They used soft K-way cluster assignments to predict each augmented view, and a contrastive loss was applied in cluster space to ensure that two views of the same patch share similar cluster assignment vectors and push different patches.

4.1.4 Cross-modal and structural matching

Contrastive learning can also go beyond simple augmentation to modalities and structural relationships beyond.

In retinal imaging, multi-modal contrastive learning has been used to align 2D fundus images with their corresponding 3D Optical Coherence Tomography (OCT) volumes from the same eye. By using Contrastive Language-Image Pre-training (CLIP) and Contrastive Leave-One-Out-Boost (CLOOB) paradigms, these frameworks map paired modalities to a shared latent space. This ensured that anatomical features are consistent across different imaging dimensions [48]. In lung ultrasound, spatial-temporal relationships were used by extracting M-mode slice images from the same grayscale B-mode video and considering them as positive pairs under a joint embedding objective [49]. LVM-Med [50] used

second-order graph matching. They formulated node matching as aligning corresponding augmented views across two batches. Instead of matching augmented pairs independently, the method also used edge and neighborhood relationships between samples, enforcing structural consistency across the augmented graphs and ensuring that similar images remain similarly connected in the latent space through second-order constraints.

These approaches move beyond simple image-level matching by using multi-modal correspondence and structural consistency into the representation space.

4.1.5 Acquisition-based and k-space contrastive learning

Unique to medical imaging acquisition, CL-MRI introduced a contrastive pretext task focused on MRI reconstruction. Instead of standard image-domain augmentations, it generates multiple views by applying different undersampling operators and acceleration factors to the same k-space data. By maximizing the mutual information between these different under-sampling levels, the model learns an under-sampling-invariant feature extractor that improves downstream reconstruction accuracy [51].

4.2 Non-contrastive learning and predictive learning

The SSL field has evolved to address the limitations of negative sampling. Non-contrastive architectures and predictive learning have gained increasing attention, because contrastive objectives may incorrectly push apart biologically similar samples. Instead, these approaches prevent representation collapse through self-distillation, redundancy reduction, or predictive modelling of intrinsic spatial and anatomical relationships.

4.2.1 Self-distillation and feature matching

Self-distillation frameworks frequently use teacher–student architectures to align different augmented views of the same input. DINO-based self-distillation has been used for dental radiograph analysis for external cervical resorption detection and dental caries classification [52] and also for periapical film segmentation in limited-label settings [53]. In these works, two augmented views of the same input image were used for the student and teacher networks. Therefore, this enforced the consistency between their embeddings.

Prior-Guided Local (PGL) extended this idea to 3D medical volumes [54]. Rather than aligning global embeddings, PGL matched voxel-level feature representations of the same local anatomical region across two transformed views. PGL used known spatial transformations to identify overlapping regions. Then, the online network predicted the aligned feature map from the target network.

This framework [55] was used for endoscopic video analysis in low-data settings. This study used Masked Siamese Networks to further adapt feature matching by predicting the prototype similarity distributions of an unmasked target view from a masked anchor view.

Vim4Path [56] applied DINO-style self-distillation to histopathology images using a Vision Mamba backbone. A student–EMA teacher framework processed multiple global and local augmented views of each whole-slide patch and aligned their output distributions without negative sampling. The learned patch embeddings were later transferred to slide-level classification via multiple instance learning. This approach encourages consistency across views while preserving long-range contextual dependencies in whole-slide images.

CADS [57] extended self-distillation to cross-modal and volumetric settings for CT segmentation. A 2D Digitally Reconstructed Radiograph (DRR) X-ray was generated from each 3D CT volume, and a transformer-based student–EMA teacher framework aligned representations between CT views and the corresponding X-ray without negative sampling. In addition, bidirectional self-distillation was enforced between corrupted CT views, and supervision was applied across multiple encoder stages.

4.2.2 Redundancy reduction

Redundancy-reduction methods use decorrelated feature dimensions across augmented views. Barlow Twins has been used to label-efficient dental caries classification using radiographs [58]. This method computes the cross-correlation matrix between the embeddings of two augmented batches and minimizes its deviation from the identity matrix. This motivated invariance between views while reducing redundancy among feature dimensions. Sliding-Window based Self-Supervised Learning applied Barlow Twins to high-resolution medical anomaly detection [59]. This network produced consistent and decorrelated embeddings for two augmented views of the same image patch extracted using a sliding window strategy.

In contrastive learning, we are using pairwise discrimination, but in this method, the representation geometry is shaped through statistical constraints.

4.2.3 Spatial and anatomical prediction

Rubik’s Cube-based methods divided 3D scans into eight equal sub-volumes and applied structured permutations and rotations (limited to horizontal and vertical directions) [60]. The model was trained to predict the type of rotation and permutation. Rubik’s Cube+ extended this by including cube masking identification [24], while volume-wise transformation strategies rotate entire anatomical layers along sagittal, coronal, or axial planes and require restoration of the original configuration [61].

Another study has done a slice-level ordering. In body part recognition using CT, the model predicted whether one slice lies anatomically above or below another [62]. Spatial awareness-based study introduced controlled corruption by inserting patches from neighboring slices and require prediction of their relative displacement (δ values) [63].

Cardiac MRI segmentation has incorporated anatomical position prediction, where 2Ch and 4Ch view planes were overlaid to define nine anatomical regions, and a model trained to classify each pixel into one of these regions (including background) [64]. In cytoarchitectonic brain analysis, Siamese networks used to predict the 3D geodesic distance between two small brain tissue patches, encouraging encoding of structural landmarks and spatial continuity [65]. These methods used intrinsic anatomical organization rather than augmentation invariance.

Med-SeAM [66], each MRI volume’s selected sagittal slices were pseudo-labeled as critical or non-critical based on anatomical position, excluding peripheral noisy slices. An intra-slice CNN backbone with cross-attention and an inter-slice BiGRU was trained to classify slice importance, encouraging the model to encode spatial dependencies and clinically relevant anatomical context across the volume.

The ASPECTS pretext task [67] leveraged atlas-based registration to generate pseudo-labels for ten clinically defined brain regions on unlabeled CT volumes. A network was then pretrained to perform multiclass segmentation of these regions. This approach forces the network to learn hemisphere-specific anatomical features and vascular-territory representations that are directly relevant to downstream stroke analysis, moving away from generic transformations toward domain-specific structural prediction.

4.2.4 Temporal and multi-modal prediction

Ultrasound video frameworks used Temporal Order Correction. First, the shuffled frames must be reordered correctly, and Spatio-Temporal Transform Prediction, where parameters of applied affine transformations (translation, scale, rotation, shear) must be identified [68]. In knee MRI analysis, jumbled patch arrangement prediction was used to learn spatial anatomical representations from video frames [69].

Another study [70] used spatial permutation prediction in a multi-modal setting. They extracted patches from different imaging modalities and shuffled them. The model then predicted a permutation matrix using the Sinkhorn operator to recover the correct spatial arrangement.

These methods have been used temporal continuity and cross-modal spatial consistency inherent in medical data.

4.3 Generative and reconstruction based learning

4.3.1 Masked image and mesh modeling

In these approaches, an encoder processes a partially masked input, and a decoder reconstructs the missing information. MAE have been applied to COVID-19 chest X-ray classification by converting images into sequences of patch tokens, masking a portion of them, and reconstructing the hidden patches using a Vision Transformer-based encoder–decoder architecture [71].

For dental panoramic radiographs, SimMIM has used masked modeling while preserving spatial structure [72]. They replaced the masked patches with a learnable token that remains in the encoder input, maintaining explicit positional information of teeth, and used a lightweight linear head to predict pixel values.

In histopathology, pyramid-based masked image modeling introduced a high masking ratio (75%) combined with wavelet-based downsampling to capture both global tissue organization and local texture patterns [73].

For CT imaging, SparK applied sparse convolutions to visible patches only and reconstructed masked regions, and this study demonstrated robustness in small-sample regimes [74].

DentalMAE applied masked mesh modeling to 3D intra-oral scans [75], which extended masked modeling to non-grid data. Non-overlapping mesh face patches were masked, and a transformer reconstructed masked patch embeddings rather than raw geometry, allowing the encoder to learn structural relationships in 3D surfaces. The masked modeling works aim to capture anatomical coherence through partial observation recovery.

Kidney tumor classification on CT images using SSL [76] used a masked autoencoder pretraining method. CT were randomly masked at the patch level and processed by a ViT encoder, while a lightweight decoder reconstructed the missing patches. In addition, a self-distillation mechanism aligned encoder and decoder outputs to transfer global semantic information into the encoder representation.

4.3.2 Context restoration and corruption recovery

Some generative approaches applied physical or structural corruptions and trained the network to restore the original image. Models Genesis introduced a unified 3D restoration objective for CT patches [77]. Random crops underwent nonlinear intensity distortion, local pixel shuffling, and in-painting or out-painting. A 3D encoder–decoder reconstructed the original patch. Similarly, a physics-inspired framework for perfusion estimation used a reconstruction method on lung-cropped 3D CT volumes [78]. The model was trained to recover original, unaugmented volumes from inputs corrupted by blur, noise, and elastic transformations, forcing the encoder to learn robust HU-density patterns and inhale–exhale functional cues.

Context restoration via patch swapping has been applied to ultrasound, CT, and MRI images [79]. Randomly exchanging non-overlapping patches disrupted spatial context while preserving intensity distribution, and the model was trained to recover the correct configuration. Inpainting-based SSL further evaluated context prediction and context restoration tasks, where random patches were zeroed or swapped before reconstruction [80].

For anomaly detection, Iterative Latent Token Masking models discrete latent tokens using a VQ-VAE and autoregressive Transformer [81]. Low-likelihood tokens were iteratively identified and masked, enabling reconstruction of a normal-data distribution even when training data contain anomalies.

Enhancing Multiple Sclerosis lesion segmentation via self-supervised pretraining [82] used a masking-based residual reconstruction objective for 3D MRI volumes. Structured brain-surface cuboids were removed and blended with blurred content to create masked inputs ((optimal mask ratio 0.4)), and a 3D VNet predicted voxel-wise residual heatmaps in a bidirectional manner. Reconstruction and anti-symmetry constraints were applied only within masked regions.

4.3.3 Cross-modal and multi-modal synthesis

In ophthalmology, multi-modal pretraining aligned retinography and fluorescein angiography images from the same eye [83, 84, 85]. These works reconstructed angiography regions of interest from retinography inputs, and this encourages learning of both shared and modality-specific retinal representations. Multimodal Image Encoding further disentangles common and exclusive feature branches during cross-modal reconstruction [86].

Auto-Generative Adversarial Network (GAN) addressed missing MRI sequences (T1, T1-contrast, T2, and T2-FLAIR) by using target-modality self-reconstruction to provide feature-level constraints [87]. A collaborative GAN generator synthesized a missing modality from available modalities. For instance, Used T1, T1-contrast (T1-C), T2 for generate T2-FLAIR.

4.3.4 Image colorization

This method is not a common method in medical image analysis because most of the image modalities are grayscale (MRI, CT, and x-rays). In endoscopic video analysis, a conditional GAN framework predicted a, b chroma channels from L luminance input in the Lab color space [88]. The method encouraged learning of semantic and texture cues in unlabeled video frames.

The Self-HER2Net framework [89] expanded this image colorization approach for breast cancer histopathology through a massive multi-task generative scheme. A network was trained to transform RGB IHC patches into multiple target representations: HSL and HSV color spaces to capture stain-intensity and membrane completeness, a DAB stain space via color deconvolution to highlight protein overexpression, and a synthetic Hematoxylin-only stain to learn nuclear morphology. By optimizing the network across these distinct color and stain domains, the framework learned complex morphology and staining patterns critical for HER2 scoring without manual labeling.

4.4 Hybrid learning

While individual pretext objectives provide distinct advantages, each exhibits inherent limitations. For instance, Generative models emphasize local reconstruction fidelity but may underemphasize global semantic invariance, and contrastive models learn strong global representations but may lack dense spatial precision. Hybrid learning co-optimized multiple objectives to use their complementary strengths within a unified training scheme.

4.4.1 Contrastive and generative integration

The DiRA framework [90] optimized jointly three objectives, which are instance or cluster-level alignment using contrastive, pixel-wise reconstruction of masked or distorted images, and a GAN-style discriminator to enforce anatomical realism. By combining these losses, the model captured both global representation structure and local anatomical detail.

MAEMC-NET [30] integrated MIM with MoCo-style contrastive learning for pulmonary nodule analysis. The framework used a ViT-Large encoder to reconstruct masked patches from axial CT slices via pixel-wise. This enforces the learning of intra-slice anatomical continuity. Simultaneously, these axial composite patches with corresponding coronal views from the same lesion, thereby enforcing cross-plane consistency and inter-lesion discrimination in the learned representation space.

Similarly, a multi-task framework [91] for coronary artery disease diagnosis in cardiac MRI combined a pixel-wise reconstruction task with two discriminative predictive tasks. The model was trained to reconstruct original images from encoded representations while simultaneously classifying whether an image was "clean vs. perturbed" by noise or "original vs. rotated". This method ensured that the network preserves dense anatomical appearance while remaining sensitive to structural orientation and noise-robustness.

Alice [92] integrated MIM using a sophisticated dual-contrastive approach. It achieved inter-volume anatomical invariance by using a pre-trained SAM landmark detector to crop and align the exact same body parts across different CT volumes, maximizing their global similarity. Simultaneously, it performed intra-volume semantic-aligned contrastive learning using a Conditional Anatomical Semantic Alignment module. The Conditional Anatomical Semantic Alignment module applied attention-based query-key matching to ensure masked and intact views are properly aligned before maximizing similarity.

For sequential medical slices, the SSMTL framework [93] used structural similarity across adjacent slices. SSMTL reconstructed two neighboring slices at a fixed distance from an anchor slice, while a Siamese-style instance discrimination branch was used for maximizing the cosine similarity between two slices from the same scan.

4.4.2 Coupling contrastive and predictive objectives

The framework for retinal disease diagnosis [94] optimized jointly rotation prediction alongside multi-view contrastive learning. The combination of this framework was designed to decouple the feature space into rotation-related structural features (capturing orientation-dependent anatomy like vessels and the optic disc) and rotation-invariant components, minimizing a combined loss to improve downstream classification.

A multi-resolution SSL framework for histopathology segmentation [95] coupled SimSiam-style representation matching with a masked-jigsaw predictive objective. They processed paired low-resolution context and high-resolution target patches from whole-slide images through separate branches, and cross-view similarity was enforced across multiple backbone stages. A Context-Target Fusion module masked and shuffled target features before concatenation with contextual representations. This motivated multi-scale structural consistency and they jointly optimized alignment and structured prediction.

4.4.3 Sequential and nested task learning

Some of the frameworks applied pretext tasks sequentially rather than co-optimizing multiple losses simultaneously. Nested Self-Supervised Learning [96] used sequential generative tasks to segment complex volumetric white matter tracts from diffusion MRI. First, they pretrained an encoder using voxel-wise streamline density regression derived from automated tractography. After that, the second pretext task initialized from the first and mimics registration-based segmentation by predicting 72 atlas-propagated tract masks, refining the network to learn tract-specific contextual anatomy prior to fine-tuning.

Table 2: Distribution of the 75 reviewed studies across SSL pretext-task families.

Pretext-task family	No. of studies
Contrastive learning	23
Non-contrastive & predictive learning	19
Generative & reconstruction-based learning	19
Hybrid learning	14
Total	75

4.4.4 Graph-based and massive multi-tasking frameworks

The framework [97] combined rotation prediction, multi-view contrastive learning, and image reconstruction for COVID-19 lesion segmentation, capturing rotation-dependent vascular structures, transformation-invariance, and segmentation-oriented localization simultaneously. Similarly, two-stage Transformers for 3D segmentation [92] combined contrastive learning on volumetric patches against in-painted negatives, masked volume inpainting to restore corrupted regions, and auxiliary rotation prediction. Early surrogate supervision framework [98] also used this multi-tasking approach, combining 3D flipping/rotation prediction, Wasserstein GAN-based 3D patch reconstruction, and conditional GAN-based image colorization.

The Self-Path framework [99] for histopathology integrated pathology-specific tasks such as multi-scale magnification prediction and a magnification-aware jigsaw puzzle with agnostic tasks like rotation, flipping, autoencoding, and gradient-reversal domain prediction to learn scale-aware morphology. For retinal images, the SSL Fuzzy Clustering Network [100] jointly optimized an autoencoder reconstruction, fuzzy C-means clustering to learn soft cluster memberships, and a SSL membership regression head. In scopy images, the ColorMe framework [101] combined pixel-wise green-red/blue colorization and red/blue color distribution estimation using Kullback-Leibler divergence pretext tasks.

A hybrid graph-level SSL framework [102] for functional MRI analysis combined denoising diffusion with contrastive alignment. It used a Markov diffusion process to progressively corrupt functional connectivity graphs and trained a Graph Transformer to reconstruct the original structure, learning diverse graph augmentations without damaging underlying Blood-Oxygen-Level-Dependent signals. This was jointly optimized with a SimSiam-style contrastive objective to maximize agreement between original and diffusion-augmented graph embeddings, enabling the encoder to learn disease-relevant brain connectivity representations without manual labels.

The UniMiSS+ framework [103] handled unpaired 2D X-rays and 3D CT volumes by combining self-distillation and generative constraints. This framework jointly pretrained a dimension-free pyramid Transformer using DINO-style view consistency, aligning 3D volumes with their extracted 2D slices (slice-volume consistency), aligning CT volumes DRRX-rays (cross-modality matching), and reconstruction.

While these categories differ in how they are formulated, their real importance depends on how well the learned representations transfer to clinical tasks. The next section systematically analyses how each pretext design aligns with downstream objectives such as classification, segmentation, detection, and registration.

5 Task alignment analysis

5.1 Distribution of downstream tasks

Table 2 reports the distribution of the 75 reviewed studies across the four pretext-task families. Contrastive learning is the most frequently adopted paradigm (23 studies), while non-contrastive and predictive learning, generative and reconstruction-based learning, and hybrid learning are represented by 19, 19, and 14 studies respectively. The relatively even spread across the latter three families indicates that the field is no longer dominated by a single paradigm, and motivates the task-alignment analysis that follows.

Among all downstream applications, classification is the most frequently studied task. It appears across a wide variety of modalities, including dermatology, chest X-ray, CT, MRI, fundus imaging, histopathology, ultrasound, dental radiography, and functional MRI. Representative classification applications include skin disease recognition, pulmonary nodule malignancy prediction, COVID-19 diagnosis and prognosis, prostate cancer diagnosis, glaucoma detection, retinal disease screening, disc degeneration grading, knee injury diagnosis, liver view classification, HER2 grading, and brain disorder detection [7, 33, 30, 34, 36, 37, 39, 40, 41, 42, 43, 52, 55, 56, 58, 59, 66, 69, 71, 73, 74, 76, 85, 86, 89, 91, 94, 97, 98, 99, 100, 102, 103]. The importance of classification reflects the importance of automated diagnostic decision support in medical AI.

Table 3: Task-alignment matrix mapping SSL pretext-task families to downstream clinical objectives. Alignment strength summarizes the predominant trend across the reviewed studies: ● strong and well-supported, ○ moderate or context-dependent, ✗ weak or prone to negative transfer.

Pretext-task family	Classification	Segmentation	Detection / anomaly	Reconstruction / regression
Contrastive (instance / patient-level)	●	○	○	✗
Contrastive (local / voxel-level)	○	●	○	○
Non-contrastive & predictive	○	●	○	○
Generative / reconstruction	○	●	●	●
Hybrid	●	●	●	○

Note: Ratings are qualitative and indicative only; alignment also depends on imaging modality (Section 5.3) and label regime (Section 5.4).

The second major downstream category is segmentation, particularly in volumetric and structurally complex medical data. These studies include such as brain tumor segmentation, multi-organ CT segmentation, cardiac MRI segmentation, retinal vessel segmentation, stroke lesion delineation, dental structure segmentation, tissue and nuclei instance segmentation in histopathology, and white matter tract segmentation [24, 35, 44, 45, 46, 53, 54, 57, 60, 61, 63, 64, 65, 67, 70, 72, 75, 77, 79, 80, 82, 83, 84, 88, 90, 92, 93, 95, 96, 97, 98, 101, 103]. Compared with classification, segmentation requires dense spatial prediction and accurate preservation of local anatomical boundaries, which makes it an especially informative testbed for evaluating representation quality.

A smaller but important subset of studies has focused on other downstream objectives. Detection-related tasks such as standard plane detection in ultrasound, anomaly detection for automated screening, object detection, and teeth numbering or restoration detection [50, 68, 72, 81, 90, 36]. Reconstruction tasks are less common but appear prominently in MRI, especially for generating high-fidelity images from undersampled k-space data [51]. Regression tasks are represented by studies on body-part position estimation, visual acuity prediction, biomarker measurement, cancer cellularity percentage, perfusion estimation, and survival prediction [32, 48, 62, 70, 38]. In addition, some papers address localization of the fovea or optic disc, saliency prediction, disease forecasting, and medical image synthesis or modality imputation, showing that SSL is increasingly being explored beyond conventional benchmarks. [48, 79, 83, 87].

Another notable observation is that several studies evaluate SSL representations across multiple downstream tasks rather than a single task only. For example, some papers jointly have reported classification and segmentation results, while others combined classification with detection, reconstruction, or localization [24, 32, 38, 50, 60, 63, 70, 77, 79, 90, 97, 98, 101, 103]. These multi-task evaluations are especially valuable because they provide evidence about whether a pretext task learns narrow task-specific features or broader transferable representations.

Overall, the downstream task distribution in the reviewed literature suggests two main points. First, SSL in medical imaging is still primarily benchmarked on classification and segmentation. Second, the increasing presence of multi-task and non-standard downstream evaluations highlights that medical SSL must support a diverse set of representation requirements. This motivates a closer analysis of how well different pretext task families align with different downstream objectives.

5.2 Alignment Between pretext tasks and downstream tasks

An important aspect of SSL in medical imaging is the relationship between the pretext task used during pre-training and the downstream task used for evaluation. The reviewed studies demonstrate that the effectiveness of SSL often depends on how well the pretext task captures features relevant to the downstream clinical objective.

Across the 75 reviewed studies, different categories of pretext tasks show different strengths depending on the type of downstream task.

Table 3 consolidates this relationship into a task-alignment matrix, mapping each pretext-task family to the four principal downstream objectives considered in this review. The ratings are not absolute scores but a qualitative synthesis of the predominant trend observed across the reviewed studies, and should be read alongside the modality and label-regime effects discussed in Sections 5.3 and 5.4. The matrix shows that no single family dominates across all objectives: contrastive learning aligns most strongly with classification, spatial and generative objectives with dense prediction tasks such as segmentation, and hybrid methods offer the most balanced profile at the cost of greater training complexity. The following subsections examine each of these alignment patterns in turn.

5.2.1 Contrastive learning and classification tasks

Contrastive learning approaches are most frequently evaluated on classification tasks, where the goal is to distinguish between disease categories or clinical conditions. Many studies using instance-level or patient-level contrastive learning demonstrate that strong performance has improved for classification problems such as disease diagnosis and screening.

For example, contrastive SSL frameworks have been applied successfully to dermatology classification, chest X-ray disease prediction, retinal disease diagnosis, neurodegenerative disorder detection, and ultrasound image classification [7, 33, 36, 37, 39, 40, 41, 49]. These methods typically learn discriminative representations by encouraging images from the same instance or patient to have similar embeddings while separating unrelated samples.

Patient-aware contrastive strategies have further improved alignment with clinical tasks by incorporating relationships between images from the same patient. Studies such as MICLe and Self-FI demonstrate that using patient-level information during pretraining can improve diagnostic accuracy and generalization [7, 41, 43]. Furthermore, instance-level contrastive learning has proven effective for extremely few-shot regimes; for instance, Ali et al. [33] achieved 86% accuracy in liver view classification with only one labeled image per class, while de Almeida et al. [34] showed that domain-specific pretraining in biparametric prostate SSL significantly outperformed ImageNet baselines for cancer diagnosis.

Overall, these findings suggest that contrastive learning is particularly effective for classification tasks, as the representation learning objective directly promotes class separability.

5.2.2 Spatial prediction tasks and segmentation problems

Pretext tasks based on spatial prediction or anatomical reasoning are commonly associated with segmentation tasks, where models must understand spatial structure and anatomical relationships.

Several studies use spatially oriented tasks such as cube rearrangement, slice ordering, or anatomical position prediction to learn representations from volumetric medical images [24, 60, 61, 62, 64]. These approaches encourage the model to capture spatial consistency within anatomical structures.

For example, Rubik’s cube-based pretext tasks have significantly improved both hemorrhage classification and brain tumor segmentation performance [24, 60], while anatomical position prediction has improved cardiac MRI segmentation in low-label scenarios [64]. Similarly, spatial-awareness-based SSL methods have improved segmentation of organs and tumors in CT datasets [63]. Specialized spatial tasks like the ASPECTS pretext by Moreau et al. [67], which used atlas-based registration to generate brain-region labels and have shown improved stability for low-contrast stroke lesion segmentation. In the domain of digital histology, Spitzer et al. [65] demonstrated that Siamese networks trained to predict 3D geodesic distance align effectively with cytoarchitectonic brain area parcellation.

These results indicate that spatial pretext tasks align well with segmentation objectives, as they force the network to learn structural relationships that are critical for delineating anatomical regions.

5.2.3 Generative and reconstruction tasks for dense prediction

Generative SSL methods, such as MIM and context restoration, are frequently used in tasks requiring dense pixel-level predictions, including segmentation and detection.

MAE approaches have reconstructed missing image regions, encouraging models to capture detailed local features. Studies using MAE-style pretraining report improved performance for tasks such as COVID-19 diagnosis, dental detection, and histopathology classification [71, 72, 73].

Similarly, context restoration tasks, where corrupted image regions must be reconstructed, have been shown to improve segmentation performance in several medical imaging applications [77, 79, 80]. The alignment of generative tasks also extends to anomaly detection; for instance, the iterative latent token masking approach by Patel et al. [81] allows models to learn the distribution of "normal" data, aligning with the detection of pathological uptake in PET scans. Furthermore, reconstruction-based tasks have shown unique alignment with MRI physics; the CL-MRI framework by Ekanayake et al. [51] used k-space contrastive learning to specifically improve downstream undersampled MRI reconstruction. In dental imaging, Almalki and Latecki [75] successfully aligned SimMIM with both teeth numbering and restoration detection.

These findings suggest that generative SSL approaches help models learn fine-grained structural features, which are particularly beneficial for segmentation and dense prediction tasks.

5.2.4 Multi-modal and cross-modal learning for complex clinical tasks

Some studies have explored cross-modal or multi-modal pretext tasks, where relationships between different imaging modalities are used for SSL.

For example, several works have learned representations by predicting or aligning information between fundus images and OCT scans, or between CT and X-ray modalities [48, 57, 83, 84]. These approaches capture complementary information across modalities and can improve performance in classification, segmentation, and regression tasks. Specific alignment is seen in retinal imaging where works by Hervella et al. [83, 85, 86] and Morano et al. [84] show that synthesizing Fluorescein Angiography from color retinography allows models to learn vascular anatomy, significantly improving vessel segmentation and diabetic retinopathy grading. Additionally, cross-modal alignment between 3D CT and synthesized 2D X-rays (DRRs) in the CADs framework [57] has been shown to improve 3D kidney and liver segmentation while significantly reducing pretraining time.

Cross-modal learning is particularly useful in medical imaging because clinical diagnosis often relies on multiple imaging modalities, and such methods encourage models to learn modality-invariant representations.

5.2.5 Hybrid approaches for generalizable representations

Recent studies increasingly combine multiple SSL objectives to create hybrid frameworks that integrate contrastive, predictive, and generative learning.

Hybrid approaches such as DiRA, UniMiSS+, and multi-task SSL frameworks demonstrate strong performance across a wide range of downstream tasks, including classification, segmentation, and detection [90, 97, 103]. These frameworks often achieve better performance than single-task SSL methods because they capture complementary aspects of the data. For example, the DiRA framework [90] has demonstrated that combining discriminative, restorative, and adversarial objectives provides massive performance boosts in highly restricted 1% label regimes for both X-ray and CT tasks. Other hybrid models like MAEMC-NET [30] combined generative and contrastive learning specifically to align axial and coronal CT views for pulmonary nodule analysis. In sequential imaging, the SSMTL framework [93] aligned adjacent MRI/CT slices to improve 3D cardiac and brain tumor segmentation. Massive multi-tasking frameworks, such as Surrogate Supervision [98] and Self-Path [99], further have demonstrated that combining pathology specific tasks like magnification prediction with generic tasks leads to superior domain adaptation in histopathology.

Such results indicate that combining multiple pretext tasks may produce more generalizable representations, especially for complex medical imaging problems.

Overall, the reviewed studies suggest that the alignment between pretext tasks and downstream tasks is not uniform. Contrastive methods are most commonly associated with classification, especially when global semantic discrimination is important [7, 33, 34, 36, 37, 39, 40, 41, 43]. In contrast, spatial prediction and anatomy-aware tasks appear more naturally aligned with segmentation problems, particularly in volumetric CT and MRI settings [24, 60, 61, 63, 64, 67]. Generative and reconstruction-based methods often provide strong performance for dense prediction and structure-sensitive tasks by encouraging the model to preserve fine local details [72, 75, 77, 79, 80, 83, 84]. Hybrid approaches extend this further by combining complementary objectives and therefore tend to perform well across multiple downstream tasks [90, 92, 93, 97, 103]. These patterns indicate that task alignment depends not only on the SSL family, but also on the type of information required by the downstream objective.

5.3 Effect of imaging modality

The effectiveness of SSL in medical imaging is also strongly influenced by the data modality. Different imaging modalities contain different types of information, such as global semantic patterns, local texture, anatomical structure, temporal continuity, or cross-modal relationships. As a result, the same pretext task may not perform equally well across all modalities.

A clear pattern in the reviewed studies is that volumetric modalities such as CT and MRI frequently benefit from pretext tasks that preserve spatial structure and anatomical consistency. Many CT and MRI based studies have used spatial prediction, anatomical ordering, reconstruction, or hybrid objectives to learn representations that are more suitable for dense prediction and volumetric analysis [24, 30, 34, 44, 46, 51, 54, 57, 60, 61, 63, 64, 67, 70, 75, 77, 80, 90, 92, 93, 96, 103]. This is especially visible in segmentation tasks, where models must capture organ boundaries, lesion extent, and inter-slice consistency. For example, Rubik’s Cube-based learning, spatial awareness tasks, and anatomical position prediction all show strong performance in CT and MRI segmentation settings [24, 60, 61, 63, 64]. Similarly, MRI reconstruction benefits from contrastive learning directly applied in k-space or acquisition-aware settings, which are naturally specific to MRI data [51].

In CT imaging, structural information and 3D spatial continuity appear particularly important. Several CT-based studies show that anatomically meaningful pretext tasks have improved performance in organ segmentation, tumor delineation, anomaly localization, and classification [30, 37, 44, 46, 54, 57, 60, 63, 67, 72, 74, 76, 77, 79, 81, 90, 98, 103]. These results suggest that CT benefits from SSL strategies that capture local context, volumetric organization, and disease-specific structural patterns. In some cases, the pretext task itself had to be adapted to the modality. For example, lower masking ratios were found to be more suitable for dental and CT-related applications because small pathological structures are easily lost when too much information is removed [30, 72, 75].

For MRI, the pattern is similar in terms of anatomical sensitivity, but MRI also appears to benefit more from methods that can handle multi-sequence or multi-view relationships. This is especially relevant in brain MRI, cardiac MRI, prostate MRI, and musculoskeletal MRI [24, 34, 35, 39, 51, 60, 64, 66, 69, 70, 78, 87, 91, 92, 93, 96]. In these settings, pretext tasks that use anatomical position, inter-slice relationships, temporal ordering, or multi-modal sequence alignment often produce stronger transfer performance. For example, anatomical position prediction improved cardiac MRI segmentation in very low-label settings [64], while multi-modal SSL across MRI sequences improved tumor and prostate segmentation as well as survival prediction [70]. These findings indicate that MRI requires SSL methods that can capture both structural and cross-sequence information.

A different trend is observed in histopathology, where local texture, stain variation, and multi-resolution tissue patterns become more important than volumetric spatial continuity. Histopathology studies are dominated by contrastive learning, self-distillation, multi-resolution learning, and pathology-specific hybrid pretext tasks [32, 47, 56, 73, 82, 89, 95, 99]. In this modality, SSL methods often focus on patch-level discrimination, slide-level aggregation, stain-related variation, or task-specific tissue structure. For example, SimCLR-based and DINO-based approaches improved both classification and segmentation in digital pathology [32, 56], while pathology-specific tasks such as JigMag showed especially strong gains in low-label regimes and domain adaptation settings [99]. This suggests that histopathology benefits from pretext tasks that preserve morphology, scale variation, and tissue-level semantics rather than purely global image discrimination.

In retinal imaging, the literature shows a strong role for cross-modal and multi-modal SSL. Many retinal studies make use of relationships between fundus photography, fluorescein angiography, and OCT data [40, 41, 43, 48, 83, 84, 85, 86, 94, 100]. This is different from CT or MRI, where the main focus is often anatomical structure within the same modality. In retinal tasks, multi-modal SSL has helped the model learn shared disease-relevant representations across imaging types. Several studies show that cross-modal reconstruction or alignment improves glaucoma detection, diabetic retinopathy grading, vessel segmentation, and prognosis prediction [48, 83, 84, 85, 86]. These findings suggest that retinal imaging is particularly suitable for multi-modal SSL because clinically meaningful information is often distributed across complementary image sources.

Ultrasound and endoscopic video modalities show another distinct pattern. Compared with CT, MRI, or histopathology, these modalities often contain temporal dynamics, operator dependence, motion variation, and lower visual consistency. As a result, SSL methods for these data often use temporal ordering, frame consistency, self-distillation, or video-based representation learning [33, 49, 55, 68, 88]. For example, SSL improved standard plane detection in fetal ultrasound and phase recognition in endoscopic video, especially when labeled data were limited [55, 68]. This suggests that in video-based or highly variable modalities, SSL methods that model temporal structure or sequence coherence are more naturally aligned than static image-based objectives alone.

Overall, the reviewed studies show that imaging modality plays a major role in determining which SSL strategy is most effective. CT and MRI benefit strongly from spatially aware and structure-preserving objectives, histopathology benefits from patch-level, multi-scale, and morphology-sensitive learning, retinal imaging benefits from multi-modal alignment, and ultrasound or endoscopic video benefits from temporal and sequential representation learning. The effect of SSL is not only task-dependent, but also modality-dependent. A pretext task that works well for one modality may not transfer equally well to another because the underlying image characteristics, structural priors, and clinically relevant features are different. This further supports the need for modality-aware design when selecting self-supervised learning strategies for medical imaging applications. Therefore, task alignment in medical SSL should not be viewed only in terms of downstream task type, but also in relation to the modality-specific characteristics of the data.

5.4 Influence of label regime

Another key factor influencing the effectiveness of SSL in medical imaging is the label regime under which downstream models are trained. Across the reviewed studies, SSL is consistently shown to provide benefit when labeled data are limited. This is particularly relevant in medical imaging, where obtaining annotations is expensive and often requires expert clinicians.

Many studies demonstrate that SSL significantly improves performance in extremely low-label or few-shot settings. For example, contrastive learning approaches have shown strong results even when only a handful of labeled examples are available. In ultrasound liver-view classification, a SimCLR-based model achieved 86% accuracy using only one labeled image per class, and reached over 98% accuracy with small labeled subsets, clearly outperforming supervised baselines [33]. Similarly, contrastive learning combined with few-shot classification enabled COVID-19 diagnosis from CT scans with strong performance even in one-shot and five-shot learning scenarios [37]. These results highlight how SSL representations capture useful structural features before any task-specific supervision is applied.

Several studies also show that SSL models can achieve comparable performance using significantly fewer labels than fully supervised training. For instance, the Rubik’s Cube-based SSL method demonstrated that training with 50% of labeled data after SSL pretraining could match the performance of models trained from scratch using 100% labels [24]. Similar findings appear in segmentation tasks, where SSL initialization allows models to maintain strong performance with reduced annotations [61, 64, 79]. In cardiac MRI segmentation, SSL improved Dice scores substantially when only a small number of annotated subjects were available, demonstrating that representation learning from unlabeled data can compensate for annotation scarcity [64].

The benefits of SSL are particularly evident in label-efficient segmentation tasks, which are common in medical imaging. Several CT and MRI segmentation studies report large performance improvements when label availability is restricted. For example, contrastive learning improved Dice scores for necrotic lung lesion segmentation and achieved performance with 10% labeled data that surpassed fully supervised training using the full dataset [46]. Similarly, context-restoration SSL improved brain tumor segmentation performance when only 25–50% of labels were available, matching the accuracy of models trained with full supervision [79]. These results indicate that SSL helps models learn spatial and anatomical priors from large unlabeled datasets.

Some studies further demonstrate that SSL can remain effective even under extremely scarce annotation regimes. In diffusion MRI tract segmentation, nested SSL achieved strong segmentation accuracy using as few as 2 to 5 labeled scans, while supervised baselines failed under the same conditions [96]. Likewise, retinal vessel segmentation models pretrained using multi-modal SSL achieved high performance even when trained with only one or two labeled images, highlighting the potential of SSL for highly annotation-limited medical tasks [83, 84].

Beyond low-label scenarios, several works have shown that SSL can still provide benefits under moderate or full-label training regimes. In large-scale histopathology studies, SSL features outperform ImageNet pretraining even when full labeled datasets are available [32]. Similarly, MAE-based methods and hybrid SSL frameworks consistently improve classification and segmentation performance when fine-tuned using full supervision [71, 90, 103]. However, the relative advantage of SSL tends to be smaller in these settings because the model can already learn effective representations directly from labeled data.

Finally, a small number of studies have explored fully unsupervised or annotation-free scenarios, where SSL representations are used directly without downstream supervision. For example, contrastive clustering in histopathology achieved high tissue classification accuracy without using any labeled data [47], while fuzzy clustering methods demonstrated strong retinal disease detection performance in a fully unsupervised setting [100]. Although these approaches are less common, they demonstrate the potential for SSL to support automated discovery of disease patterns without manual annotation.

Overall, the reviewed literature shows that the impact of SSL is strongly dependent on the label regime. SSL provides the most substantial benefits in low-label and few-shot scenarios, where learned representations significantly reduce the reliance on expensive medical annotations. As label availability increases, SSL still improves performance and robustness, but the magnitude of improvement becomes smaller. These findings highlight the importance of SSL as a key strategy for addressing the fundamental challenge of limited labeled data in medical imaging.

5.5 Positive and negative transfer

The SSL aims to learn generalizable representations from unlabeled data that can be transferred to downstream tasks. However, the effectiveness of this transfer is highly dependent on the alignment between the pretext task, the data distribution, and the downstream task requirements. Across the reviewed literature, most studies report strong positive transfer, where SSL pretraining significantly improves downstream performance. Nevertheless, several studies also highlight cases of limited or negative transfer, where the benefits of SSL are reduced or where inappropriate pretext tasks lead to weaker representations.

5.5.1 Positive transfer

The majority of reviewed studies demonstrate clear positive transfer, where SSL-pretrained models outperform models trained from scratch or models initialized with generic pretraining such as ImageNet. These improvements are particularly evident in medical imaging tasks where labeled data are scarce.

For example, contrastive learning methods frequently lead to strong performance gains across classification tasks. Multi-instance contrastive learning improved dermatology classification accuracy by 6.7% and increased chest X-ray diagnosis Area Under Curve (AUC) compared to supervised ImageNet pretraining [7]. Similarly, SimCLR-based SSL significantly improved ultrasound liver-view classification, achieving 98.7% accuracy and outperforming fully supervised baselines even with extremely limited labeled data [33].

Positive transfer is also widely observed in segmentation tasks. Spatial and anatomical pretext tasks such as Rubik’s Cube prediction improved both classification and segmentation performance in 3D medical imaging, increasing classification accuracy by 8.11% and segmentation Dice scores by more than 8% compared with training from scratch [24]. Similar improvements were reported in multi-organ CT segmentation, where SSL pretraining consistently improved Dice scores across multiple organs and tumor types [54]. Large-scale pretraining frameworks such as Models Genesis further demonstrated consistent improvements across multiple modalities including CT, MRI, and X-ray [77].

Several studies also report strong transferability across datasets and modalities. For instance, histopathology models trained with SimCLR representations significantly outperformed ImageNet features across multiple cancer classification tasks, with average F1 improvements exceeding 28% in linear evaluation settings [32]. Similarly, universal medical SSL frameworks trained on large multi-modal datasets demonstrated strong transferability across both 2D and 3D medical imaging tasks, outperforming several baseline SSL methods across segmentation and classification benchmarks [103].

In low-label regimes, positive transfer becomes even more pronounced. SSL representations allow models to achieve competitive performance with significantly fewer annotations. In retinal vessel segmentation, SSL pretraining enabled strong performance even when only one or two labeled images were available [83], [84]. Similarly, nested SSL enabled accurate white matter tract segmentation using only 2–5 labeled scans, outperforming baseline methods by a large margin [96]. These findings demonstrate that SSL can substantially reduce the dependency on costly medical annotations.

5.5.2 Negative or limited transfer

Despite the widespread success of SSL, several studies highlight situations where transfer performance is limited or where pretext tasks may introduce undesirable biases. One common issue arises when the pretext task is poorly aligned with the downstream objective. For example, in contrastive multi-modal retinal analysis, simply increasing the amount of data by concatenating different modalities actually reduced performance by approximately 6%, indicating that naive multi-modal fusion may harm representation learning when modality relationships are not properly modeled [43].

Another limitation occurs when SSL representations are overly specialized to specific anatomical structures or imaging conditions. In voxel-level contrastive learning for CT segmentation, pretrained representations performed strongly for organ segmentation tasks but showed weaker performance for certain tumor segmentation tasks when evaluated without fine-tuning [44]. This suggests that features learned from structural anatomy may not fully capture pathological variations.

Negative transfer can also arise when pretext tasks distort important medical features. For example, MIM approaches in medical imaging often require significantly lower mask ratios compared with natural images because pathological features are frequently small and localized. Studies in pulmonary CT and dental radiography show that large masking ratios can degrade downstream performance by removing clinically relevant information [30, 72].

Finally, some studies highlight that SSL benefits may diminish in highly supervised regimes where large labeled datasets are already available. In such cases, the performance gap between SSL and fully supervised training becomes smaller, as models can learn strong task-specific representations directly from labeled data [32, 38].

Overall, the reviewed literature indicates that positive transfer is the dominant outcome of SSL in medical imaging, particularly in low-label scenarios where pretrained representations significantly improve downstream performance. However, the effectiveness of transfer depends strongly on task alignment, modality characteristics, and pretext task design. Poorly aligned pretext tasks, excessive data corruption, or mismatched modalities can lead to limited or negative transfer. These observations highlight the importance of designing task-aware and modality-aware SSL frameworks for medical imaging applications.

Taken together, the analyses across task alignment, modality characteristics, label regimes, and transfer behavior show that the success of SSL in medical imaging is not solely determined by the choice of algorithm. Instead, performance

Table 4: Practical design recommendations for task-aligned SSL in medical imaging, synthesized from the reviewed literature.

Downstream goal	Recommended pretext task	Rationale	Representative studies
Classification	Instance- or patient-level contrastive learning	Promotes global semantic discriminability and class separability	Azizi et al. [7], Ali et al. [33], de Almeida et al. [34]
Segmentation	Spatial / anatomical prediction; local or voxel-level contrastive	Preserves anatomical boundaries and inter-region spatial structure	Bai et al. [64], Goncharov et al. [44], Tao et al. [61]
Detection & anomaly localisation	Masked image modelling; context restoration	Retains fine-grained local texture and contextual cues	Almalki & Latecki [72], Patel et al. [81], Zhou et al. [77]
Video / temporal analysis	Temporal-order or frame-consistency prediction	Captures motion and inter-frame anatomical transitions	Jiao et al. [68], Hirsch et al. [55]
Multimodal diagnosis	Cross-modal alignment or synthesis	Learns modality-invariant yet complementary representations	Suekei et al. [48], Hervella et al. [85], Taleb et al. [70]

depends strongly on the alignment between pretext tasks, imaging modality, and downstream objectives. Therefore, designing domain-aware pretext tasks and using modality-specific priors remains a critical direction for future SSL research in healthcare.

6 Practical design guidelines for task-aligned SSL in medical imaging

Based on the analysis of the reviewed studies, it is clear that the success of SSL in medical imaging does not depend only on the choice of algorithm, but more importantly on how well the pretext task is aligned with the downstream clinical objective, the imaging modality, and the available label regime. Across the literature, there is no evidence that SSL strategy consistently performs best for all applications. Instead, different pretext task families show different strengths depending on whether the target problem requires global semantic discrimination, dense spatial prediction, temporal reasoning, or multi-modal fusion. Therefore, the design of medical SSL frameworks should be task-aware, pathology-aware, and modality-aware rather than directly adapted from general computer vision. This section summarizes the main practical design guidelines derived from the reviewed studies and translates them into recommendations for selecting and designing SSL strategies in medical image analysis.

Table 4 consolidates the main design recommendations derived from the reviewed studies, linking each downstream goal to a recommended class of pretext task, the underlying rationale, and representative supporting studies. The remainder of this section discusses each recommendation in detail.

6.1 Select the pretext task according to the downstream objective

One of the most important considerations when designing a SSL pipeline is selecting a pretext task that aligns with the requirements of the downstream objective. The studies reviewed in this survey consistently show that different downstream tasks benefit from different types of representation learning. In other words, there is no single pretext task that works best for all medical imaging problems. Instead, the choice of SSL strategy should reflect the type of information the downstream task depends on.

For classification tasks, methods that encourage global semantic discrimination generally perform well. Contrastive learning is particularly effective in this setting because it learns representations that separate clinically different samples while remaining invariant to transformations that do not change the underlying diagnosis. This property is useful for disease diagnosis and screening, where the model needs to capture high-level visual patterns associated with different pathological conditions. Across a wide range of modalities including dermatology images, chest X-rays, retinal fundus images, ultrasound, and MRI contrastive pretraining has repeatedly been shown to improve classification performance compared with both training from scratch and ImageNet-based transfer learning [7, 33, 34, 36, 37, 39, 40, 41].

Segmentation tasks, however, place very different demands on the learned representations. Instead of global semantic discrimination, segmentation requires the model to capture detailed spatial structure, anatomical boundaries, and relationships between neighboring regions. As a result, pretext tasks that encourage spatial reasoning or anatomical awareness tend to be more beneficial. Several studies demonstrate that spatial prediction tasks such as Rubik’s Cube transformations, anatomical position prediction, and spatial awareness learning lead to improved segmentation performance across organs and tumor regions [24, 60, 61, 63, 64]. Similarly, approaches that learn local patch or

voxel-level correspondences have shown strong results in multi-organ and tumor segmentation tasks [44, 45]. These findings suggest that dense prediction tasks benefit most from pretext tasks that explicitly force the encoder to learn local structure rather than only global invariance.

For detection, anomaly localization, and structure-sensitive analysis, pretext tasks that preserve fine-grained texture and contextual information can be particularly useful. Generative or reconstruction-based approaches such as masked image modeling, context restoration, and inpainting-based learning have demonstrated strong performance in applications such as dental detection, anomaly detection, and lesion identification [59, 72, 77, 79]. In many of these cases, the downstream task depends on subtle pathological patterns, so the pretext objective should avoid discarding important local details during representation learning.

When the downstream task involves temporal reasoning, such as ultrasound video analysis, endoscopic phase recognition, or sequential MRI interpretation, the pretext design should capture temporal dependencies such as frame ordering, motion consistency, or inter-slice relationships [55, 68, 69]. Temporal SSL tasks allow the model to learn meaningful representations from sequential medical data without requiring manual annotations.

Similarly, when the target application relies on complementary information from multiple imaging modalities, cross-modal or multi-modal pretext tasks are often more appropriate than single-view objectives. These approaches encourage the model to align representations across modalities and use complementary diagnostic signals [48, 70, 83, 84, 85, 86].

Overall, the design of the pretext task should be guided by the requirements of the downstream objective. Global semantic discrimination is particularly useful for classification tasks, structural and spatial reasoning benefits segmentation, preservation of local detail is important for detection and anomaly localization, temporal modeling supports video-based analysis, and cross-modal alignment is valuable for multi-modal medical imaging applications.

6.2 Incorporate modality-specific priors into SSL design

Another important observation from the reviewed literature is that the effectiveness of SSL is strongly influenced by the characteristics of the imaging modality. Medical images differ substantially from natural images in terms of spatial structure, dimensionality, noise patterns, and acquisition physics. As a result, SSL frameworks that explicitly use modality-specific priors tend to produce more transferable representations than generic approaches directly adapted from natural image benchmarks.

For volumetric imaging modalities such as CT and MRI, SSL frameworks should explicitly incorporate volumetric and anatomical priors during pretraining. Several studies, therefore, design pretext tasks that have used volumetric structure, including slice ordering, anatomical position prediction, voxel-level contrastive learning, and spatial puzzle solving [24, 44, 60, 61, 63, 64]. These approaches motivate the model to learn global anatomical organization while also preserving local structural information, which is particularly beneficial for segmentation and structural classification tasks. The evidence shows that such anatomy-aware objectives improve performance in tumor segmentation, organ segmentation, and brain pathology classification compared with both training from scratch and standard ImageNet-based transfer learning [24, 61, 63].

For histopathology images, SSL design should consider extremely high spatial resolution and multi-scale tissue structure. In this domain, patch-based contrastive learning and multi-resolution representation learning are frequently used to capture morphological patterns at different scales [32, 56, 73, 95]. Methods that use large collections of unlabeled tissue patches or whole-slide images have demonstrated significant improvements in classification, segmentation, and regression tasks related to cancer diagnosis and tissue characterization [32, 56, 73]. These findings suggest that SSL frameworks for histopathology should explicitly account for multi-scale structures and heterogeneous tissue appearance.

For retinal imaging, SSL frameworks should use anatomical priors associated with the consistent spatial arrangement of key retinal structures. Several studies have introduced pathology-aware or multi-modal pretext tasks that encourage the model to focus on clinically meaningful regions rather than purely global image features [40, 43, 48, 83, 85, 86]. These approaches have improved disease classification accuracy and produced representations that align better with clinically interpretable structures.

For projection-based modalities such as X-ray imaging, SSL objectives should maintain subtle pathological features that may occupy a small spatial region, and global anatomical layout remains relatively stable. In chest X-ray and dental radiography, studies have shown that contrastive, redundancy-reduction, and MIM can all be effective when they preserve diagnostically relevant fine-grained structures rather than relying only on generic natural-image augmentations [36, 38, 52, 58, 71, 72]. In particular, chest X-ray studies demonstrated strong gains in performance of the disease classification, while dental radiography studies show that smaller masking ratios are often necessary because clinically important anatomical features can be easily lost when excessive information is removed during pretraining [38, 58, 71, 72].

For imaging modalities with temporal dynamics such as ultrasound videos and endoscopic recordings, SSL design should include temporal consistency and sequential reasoning. In these modalities, SSL methods that incorporate temporal coherence or sequence-based reasoning tend to perform better than static image-based objectives [55, 68, 69]. Pretext tasks such as frame ordering, temporal prediction, or sequential representation learning allow the model to capture motion patterns and anatomical transitions across frames, which has improved performance in applications such as surgical phase recognition, ultrasound plane detection, and video-based diagnosis [55, 68].

Overall, these findings highlight that the design of SSL frameworks should consider the underlying imaging physics and anatomical structure of each modality. Instead of focusing solely on generic computer vision pretext tasks, including modality-aware constraints can substantially improve representation quality and downstream task performance.

6.3 Match the SSL strategy to the available label regime

Another important design consideration in SSL is the availability of labeled data. One of the primary motivations for moving to SSL in medical imaging is the high cost and difficulty of obtaining expert annotations. As a result, the choice of SSL strategy should be aligned with the expected label regime of the downstream application. Evidence across the reviewed studies consistently shows that SSL provides the greatest benefit when labeled data are limited but large collections of unlabeled medical images are available [33, 37, 64, 96, 98].

In extremely low-label settings, such as few-shot or small-sample training, SSL pretraining often provides good performance improvements. Contrastive and reconstruction-based approaches have demonstrated strong effectiveness in these scenarios because they allow the model to learn meaningful anatomical and structural representations before fine-tuning. Several studies have shown that SSL pretrained models can achieve competitive performance using only a small fraction of labeled data compared with fully supervised models trained from scratch [33, 37, 64, 83, 84]. For example, contrastive learning approaches in ultrasound imaging and chest CT diagnosis demonstrate strong diagnostic accuracy even with very limited labeled samples, and also anatomy-aware SSL pretraining for cardiac MRI segmentation has shown significant performance gains when only a few labeled subjects are available [33, 64, 83]. These results indicate that SSL can effectively reduce the annotation burden in clinical workflows.

In moderately labeled scenarios, where a limited but non-negligible amount of annotated data is available, SSL pretraining often has improved both convergence speed and final model performance. Many studies report that models initialized with SSL representations outperform models initialized with ImageNet weights or random initialization when fine-tuned on domain-specific medical datasets [7, 34, 38, 50, 77]. This advantage is particularly pronounced for 3D medical imaging tasks such as CT and MRI, where the structural characteristics of medical images differ significantly from natural images and therefore benefit from domain-specific representation learning [77, 98].

Even when larger labeled datasets are available, SSL can still provide benefits by improving generalization and robustness. Several studies have demonstrated that SSL pretrained models have shown stronger cross-dataset transfer, better robustness to distribution shifts, and improved performance when applied to data from different institutions or acquisition protocols [7, 36, 48, 50, 103]. For example, large-scale SSL pretraining across multiple medical imaging modalities has been shown to produce representations that generalize well across diverse downstream tasks and datasets.

An additional advantage of SSL is its ability to use large collections of unlabeled medical images. Hospitals and medical centers often store extensive imaging archives that remain unused for Machine Learning (ML) due to the lack of annotations. SSL pretraining enables these unlabeled datasets to be effectively used, allowing models to learn general anatomical and structural representations before being adapted to specific diagnostic tasks [32, 50, 77]. This capability is particularly valuable for building scalable medical AI systems in data-constrained environments.

Overall, these findings suggest that SSL should be considered a primary training strategy when labeled data are scarce and unlabeled data are abundant. Designing SSL pipelines that explicitly account for the expected label regime can significantly improve model efficiency and reduce the need for costly manual annotations.

6.4 Use multi-modal and hybrid objectives when complementary signals exist

Another important guideline from the reviewed studies is the increasing effectiveness of hybrid and multi-modal SSL frameworks. Early SSL methods typically depended on a single pretext objective. However, recent studies have shown that combining multiple pretext objectives or introducing new objectives can produce richer and more transferable representations. Hybrid frameworks integrate complementary objectives, allowing the model to capture both global semantic structure and fine-grained anatomical information [30, 90, 92, 93].

Several studies have demonstrated that combining contrastive and generative objectives improves performance across a wide range of downstream tasks. Hybrid approaches typically use contrastive learning to enforce semantic separation

between samples, while reconstruction-based tasks preserve spatial and structural information. This combination helps prevent overly invariant representations that may discard clinically relevant details. Reviewed studies have shown that such hybrid objectives can outperform single-objective SSL methods in both classification and segmentation tasks across modalities, including CT, MRI, and chest X-ray imaging [30, 90, 92].

Multimodal self-supervised learning becomes particularly valuable when different imaging modalities capture complementary clinical information. For example, retinal imaging studies have combined fundus photographs with fluorescein angiography or OCT scans to learn modality-invariant representations that better capture retinal structure and pathology [83, 84, 85, 86]. Similarly, cross-modal alignment between different MRI sequences or between CT and X-ray images enables models to capture the relationships between modalities during representation learning [48, 70]. These multi-modal objectives motivate the model to learn shared anatomical features while still preserving modality-specific characteristics that may be diagnostically important.

Another trending direction is cross-dimensional representation learning, where models are trained jointly on both 2D and 3D medical data. Since medical imaging datasets often contain both slice-based images and volumetric scans, hybrid SSL frameworks can use information from both formats during pretraining. Approaches that learn shared representations across dimensions allow knowledge transfer between imaging formats and improve generalization across anatomical regions and modalities [50, 103].

Hybrid frameworks can also combine multiple pretext tasks within a single training pipeline. Some approaches use sequential or nested SSL tasks, while others optimize several objectives simultaneously during pretraining. These strategies motivate the model to learn complementary representations that capture spatial structure, semantic information, and contextual relationships within medical images [93, 96]. These results suggest that such multi-task SSL strategies can improve label efficiency and increase robustness to domain shifts across datasets.

Overall, these findings suggest that combining complementary learning objectives and using relationships across imaging modalities can substantially improve representation quality. Hybrid and multi-modal SSL frameworks therefore represent a successful direction for developing more generalizable and clinically meaningful medical image representations.

6.5 Design pretext tasks to minimize negative transfer

While SSL has demonstrated clear benefits across many medical imaging applications, the reviewed studies also indicate that poorly aligned pretext tasks can lead to negative transfer, where the learned representations fail to benefit or may even degrade downstream performance. This issue typically arises when the SSL objective motivates invariances or transformations that conflict with clinically relevant image features. Therefore, careful design of pretext tasks is necessary to ensure that the learned representations preserve diagnostically meaningful information [30, 77, 98].

One common source of negative transfer occurs when augmentation strategies remove subtle pathological patterns. In many medical images, clinically important signals occupy only small or localized regions, such as micro-calcifications in mammography, small tumors in CT scans, or fine vascular structures in retinal images. If the pretext task motivates excessive invariance to these structures, the encoder may learn representations that ignore critical diagnostic cues. Several studies therefore report improved performance when augmentation policies are adapted to medical imaging characteristics rather than directly adopting augmentation strategies developed for natural images [40, 43, 58].

Masking-based generative methods further highlight the importance of task alignment. In natural image applications, MAE commonly uses high masking ratios to encourage strong contextual reasoning. However, multiple medical imaging studies have reported that lower masking ratios are often more suitable because anatomical information is densely distributed across the image. For example, studies using MAE frameworks for chest X-ray and dental radiographs have shown that reducing the masking ratio leads to better downstream classification and detection performance [71, 72]. Excessive masking can make the reconstruction task overly difficult and may force the model to hallucinate anatomical structures rather than learn meaningful representations.

Negative transfer can also arise when the pretext task fails to capture spatial or structural properties required by the downstream objective. For example, global instance-level discrimination may not adequately support segmentation tasks that depend on accurate boundary localization and spatial relationships. In such cases, spatially aware pretext tasks, including patch-level prediction, voxel-level contrastive learning, or anatomical position prediction, have been shown to produce more useful representations for dense prediction problems [44, 60, 63, 64]. Similarly, studies that explicitly model temporal dynamics and multi-modal relationships demonstrate improved performance in video-based and multi-modal imaging tasks [68, 70], suggesting that ignoring these relationships may limit representation quality.

To reduce the risk of negative transfer, SSL frameworks should therefore include domain-aware transformations and task-specific constraints. Designing pretext tasks that align with anatomical structure, imaging physics, and clinical

objectives can significantly improve the transferability of learned representations. Additionally, evaluating pretrained representations across multiple downstream tasks and datasets can help verify that the learned features generalize effectively rather than overfitting to the specific pretraining objective. Several studies have demonstrated that domain-specific SSL objectives and appropriately designed pretraining strategies significantly improve downstream performance in medical imaging applications [30, 77, 103].

Overall, these findings highlight that effective SSL design requires not only selecting powerful pretext tasks but also ensuring that these tasks remain compatible with the characteristics of medical imaging data. Avoiding negative transfer is important for developing robust representations that can support a wide range of downstream diagnostic and analytical applications.

7 Open challenges and limitations

Despite the rapid growth of SSL in medical imaging, several open challenges and limitations remain that hinder its widespread adoption in clinical applications. Addressing these challenges and limitations is important.

One of the challenges in SSL for medical image analysis, identified in the reviewed studies, is its limited generalizability across medical distributions. Most SSL methods are evaluated on a limited number of data points, and often those data are collected from a single institution or under similar acquisition protocols. The main reason for the limited number of data is that most medical data is not publicly available due to ethical and cost constraints. An important point is that medical images are highly sensitive to variations in scanner type, imaging parameters, patient populations, and clinical workflows. As a result, models trained on data from one distribution may degrade in performance when applied to images from a different distribution. However, most studies have reported greater robustness than fully supervised models; systematic evaluation across different data distributions remains limited. Therefore, ensuring consistent performance across institutions remains an important open challenge for deploying SSL based medical imaging systems in real-world settings.

Another important challenge arises from potential misalignment between the pretext task used in SSL and downstream clinical objectives. SSL models learn representations by optimizing the pretext tasks. However, these tasks do not always reflect the characteristics of the final diagnostic task. When the pretext task motivates invariances that suppress clinically relevant features, the learned representations may lead to negative transfer during downstream training. For example, in multi-modal learning settings, combining multiple modalities without explicitly modeling their relationship can degrade performance, and studies have reported decreases of approximately 5–6% when modality interactions are poorly captured. These observations highlight the importance of carefully designing pretext tasks aligning with the anatomical, spatial, or modality-specific properties required for the target clinical application.

In medical imaging, annotation is limited and costly, SSL is often promoted as a solution for this. However, its effectiveness still depends on the label regime of the downstream task. Some of the studies have reported results gained in extremely low-label settings where SSL method significantly improves performance compared with training from scratch. However, when the number of labeled data increases, the benefits of the SSL may become smaller and inconsistent across tasks. Sometimes, when models are trained with sufficient labeled data, they may approach the performance of SSL models. Therefore, this dependence on the label regime makes it difficult to determine when SSL provides the benefit and highlights the need for more systematic evaluations across different label availability scenarios.

It might be difficult to isolate the true contributions of SSL from other factors that influence model performance. Many recent SSL studies use increasingly powerful architectures or large-scale pretrained backbones. In such scenarios, downstream performance improvements may come not only from SSL objectives, but also from architectural advances, larger model capacity, or improved fine-tuning strategies. It has become challenging to determine whether performance gains are primarily because of SSL framework itself, because all these factors are often introduced simultaneously. Therefore, standardized evaluation protocols are needed to better understand the independent impact of SSL on medical image representation quality.

Another practical limitation of SSL is the high computational cost with large-scale pretraining. Some of the SSL approaches may depend on the large batch sizes, long training schedules, and high-capacity architectures. Therefore, training such models often requires multiple Graphics Processing Units (GPUs) and extensive computational resources, which may not be readily available in many clinical research environments. In addition, medical images are high resolution (histopathology) or volumetric data (CT, MRI). This further increases memory and computational requirements during pretraining. Therefore, developing more computationally efficient SSL strategies remains an important direction for enabling broader adoption of SSL in medical imaging.

8 Future research directions

One of the limitations of current SSL frameworks in medical imaging is the generalizability of models trained on specific available data. Therefore, future research should explore decentralized SSL frameworks that enable multi-institutional pretraining without requiring direct data sharing. Such approaches could help SSL models to use a large collection of medical images. In addition, developing acquisition-invariant pretext tasks that focus on anatomical structures rather than scanner types or imaging parameters may further improve the robustness of SSL representations across heterogeneous imaging distributions.

Another important direction is the development of pathology-aware pretext tasks that explicitly prioritize subtle diagnostic features. Unlike generic pretext tasks, these approaches aim to guide SSL models toward clinically meaningful patterns, which are often sparse and difficult to capture. Designing such tasks could significantly enhance the clinical relevance and diagnostic utility of learned representations.

Another direction involves improving the alignment between pretext tasks and downstream clinical objectives, and also, currently, there is no universally optimal objective for all medical imaging applications. Therefore, automated strategies for task selection and optimization can be investigated. For example, hybrid SSL frameworks could use meta-learning or adaptive objective weighting to dynamically balance different learning signals, depending on the target clinical task. In addition, developing pathology-aware pretext tasks that prioritize subtle diagnostic features represents an important opportunity for advancing SSL in medical imaging.

Another direction is that the computation cost of SSL pretraining, which remains a barrier for widespread adoption in medical imaging research. Modalities such as CT and MRI, as well as high-resolution images such as histopathology, often require substantial memory resources and extended training times. Therefore, future research can focus on developing resource-efficient SSL frameworks capable of learning robust representations from high-resolution or volumetric medical data. To achieve this, memory-efficient architectures, lightweight representation-learning strategies, and optimized training schemes for 3D medical images would be a good direction. In addition, exploring adaptive masking strategies or sparsity-aware learning methods may help models focus on informative anatomical regions while reducing redundant computation.

Another direction is improving evaluation methodologies for SSL in medical imaging. Many recent studies combine SSL pretraining with powerful model architectures, large transformers, or advanced fine-tuning strategies. As a result, it can be difficult to isolate the true contribution of the SSL objective itself. Therefore, future research focuses on developing standardized evaluation protocols that allow fair comparison between SSL methods and supervised baselines. In addition, future evaluations need to be extended beyond traditional metrics to include measures of robustness, out-of-distribution generalization, and model interpretability.

9 Conclusion

This systematic review has provided a task-oriented analysis of SSL in medical imaging, with a particular focus on the relationship between pretext task design and downstream clinical objectives. By categorizing 75 studies into contrastive, non-contrastive, generative, and hybrid learning paradigms, we examined how different supervisory signals influence representation learning and transfer performance.

Our findings emphasize that the effectiveness of SSL is highly context-dependent. There is no universally optimal pretext task; instead, performance is governed by the alignment between the pretraining objective, the target clinical task, and the imaging modality. Contrastive methods are effective for learning global discriminative representations, particularly for classification tasks, but their performance may be limited in scenarios involving subtle or localized pathology. In contrast, generative and spatial prediction-based approaches better preserve anatomical structure and are more suitable for segmentation and dense prediction tasks. Non-contrastive and hybrid methods further demonstrate strong and stable performance across diverse settings.

In addition, modality-specific characteristics play a critical role in SSL design. Volumetric modalities such as CT and MRI benefit from spatially aware and 3D-consistent objectives, while histopathology requires multi-scale and morphology-sensitive representations. Importantly, SSL shows the greatest advantage in low-label and few-shot regimes, helping to alleviate the reliance on costly expert annotations.

Despite these advances, several challenges remain. Future work should focus on developing pathology-aware and clinically aligned pretext tasks, improving computational efficiency for high-resolution and volumetric data, and establishing standardized evaluation protocols that enable fair comparison across methods.

Overall, advancing SSL in medical imaging requires a shift from generic representation learning toward domain-aware, task-specific, and clinically meaningful design. Such an approach will be essential for translating SSL methods from research settings into reliable and generalizable clinical applications.

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